

ML- LAB - SVM Classifier Lab

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Moon Data Set :

```
... SVM with LINEAR Kernel <PES2UG23CS333>
      precision    recall  f1-score   support

         0         0.85         0.89         0.87         75
         1         0.89         0.84         0.86         75

    accuracy          0.87         150
  macro avg          0.87         150
weighted avg          0.87         150
```

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SVM with RBF Kernel <PES2UG23CS333>
      precision    recall  f1-score   support

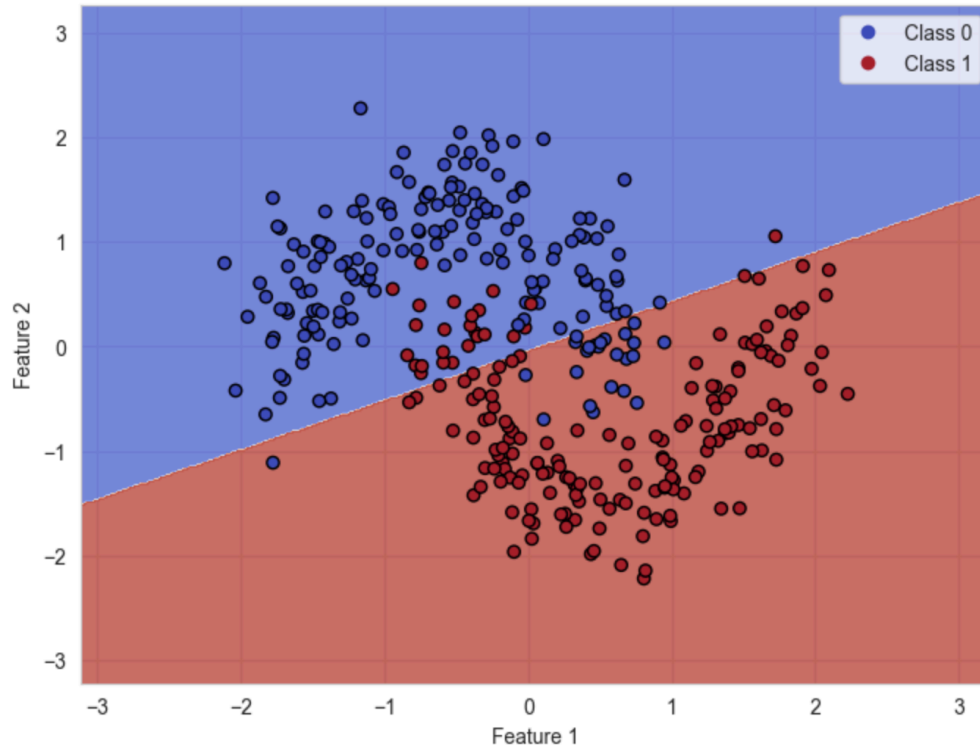
         0         0.95         1.00         0.97         75
         1         1.00         0.95         0.97         75

    accuracy          0.97         150
  macro avg          0.97         150
weighted avg          0.97         150
```

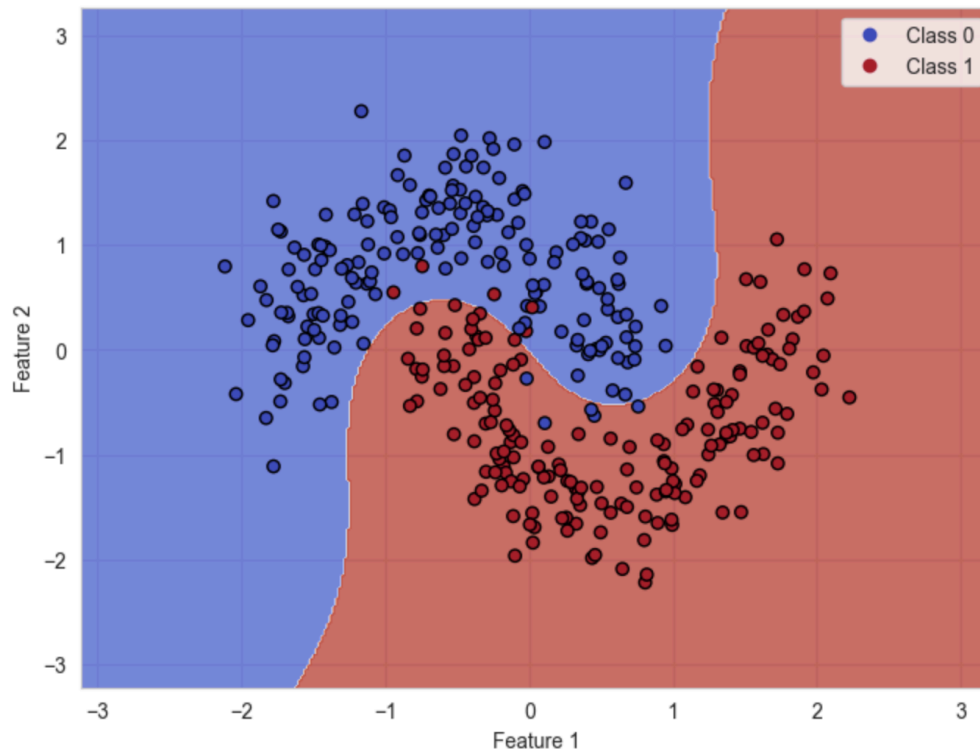
```
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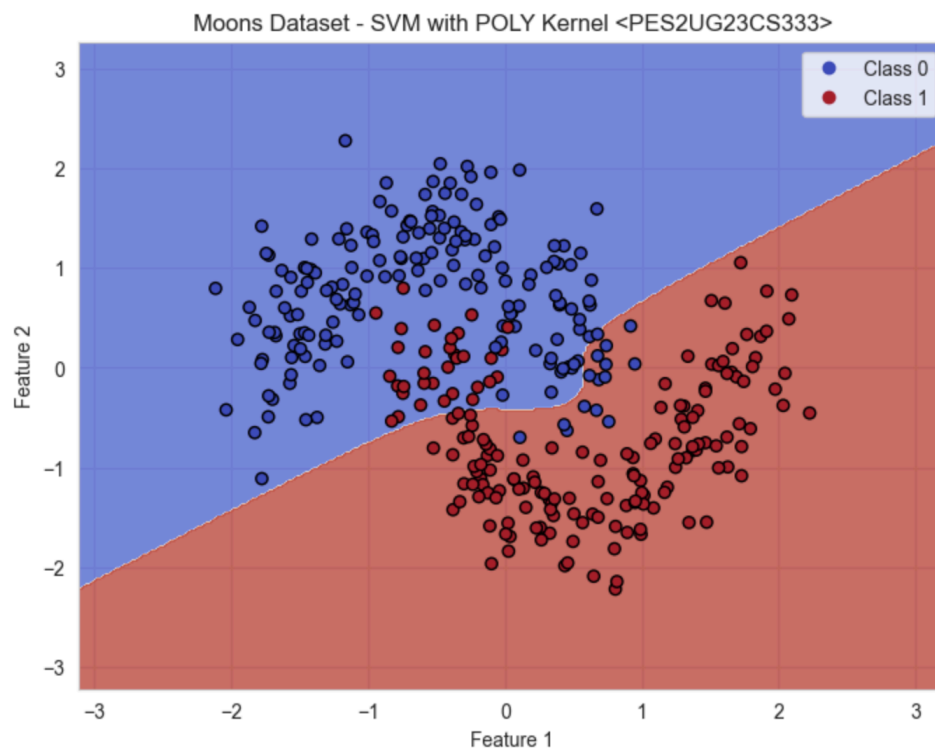
SVM with POLY Kernel <PES2UG23CS333>
...
weighted avg          0.89         0.89         0.89         150
```

Moons Dataset - SVM with LINEAR Kernel <PES2UG23CS333>



Moons Dataset - SVM with RBF Kernel <PES2UG23CS333>





Banknote Authentication

```
... SVM with LINEAR Kernel <PES2UG23CS333>
      precision    recall  f1-score   support

   Forged         0.90      0.88      0.89         229
   Genuine         0.86      0.88      0.87         183

 accuracy          0.88          0.88         412
 macro avg         0.88      0.88      0.88         412
 weighted avg      0.88      0.88      0.88         412

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SVM with RBF Kernel <PES2UG23CS333>
      precision    recall  f1-score   support

   Forged         0.96      0.91      0.94         229
   Genuine         0.90      0.96      0.93         183

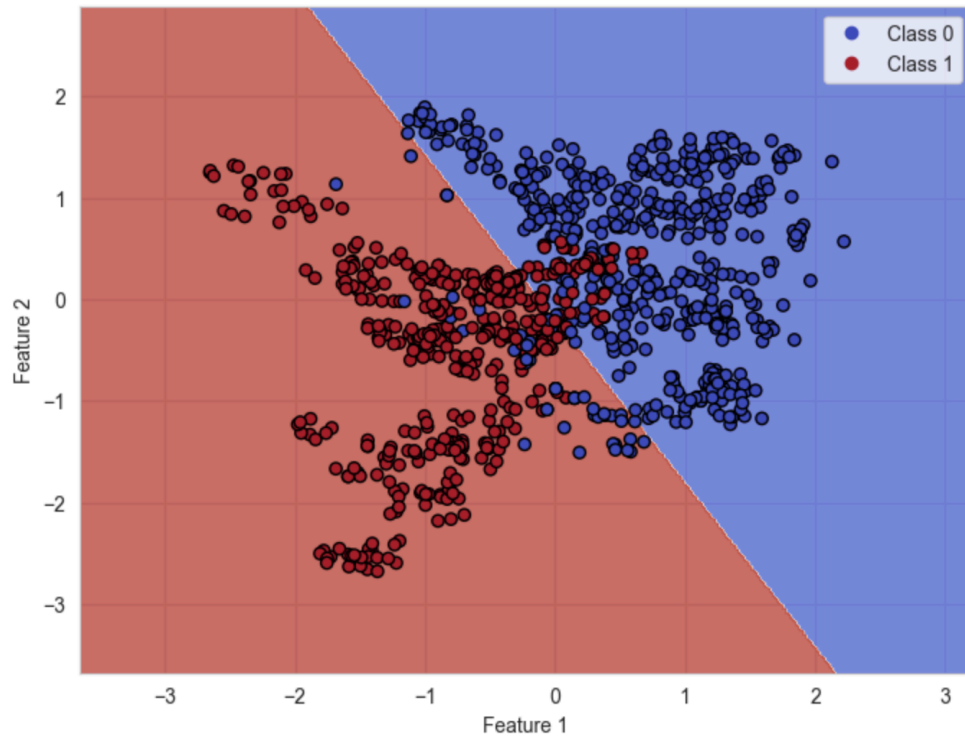
 accuracy          0.93          0.93         412
 macro avg         0.93      0.93      0.93         412
 weighted avg      0.93      0.93      0.93         412

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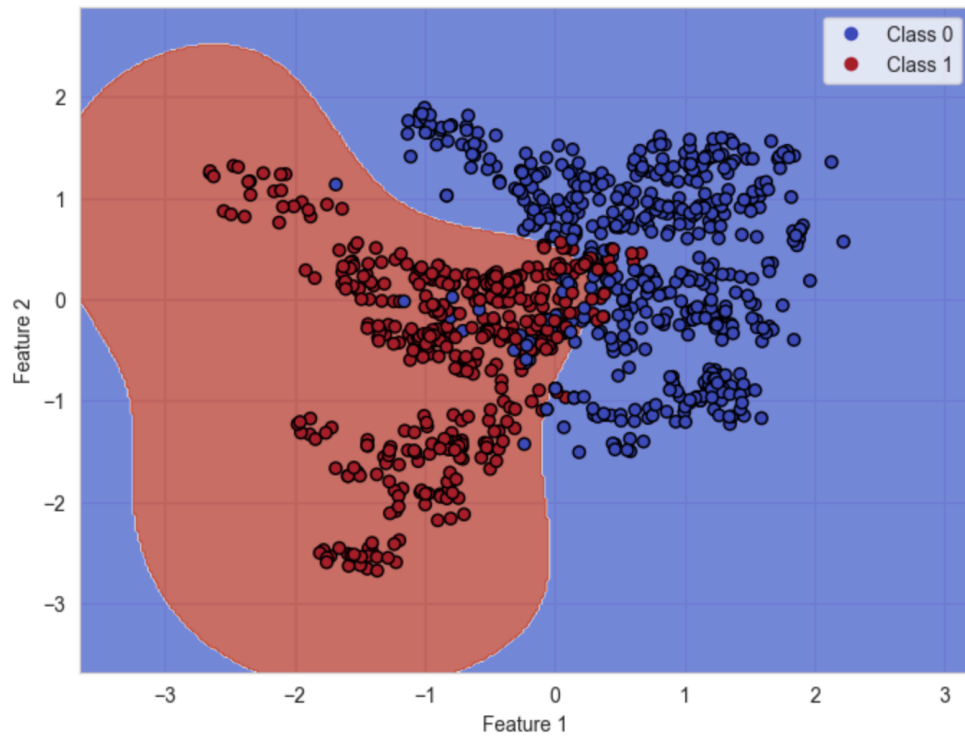
SVM with POLY Kernel <PES2UG23CS333>
...
weighted avg      0.85      0.84      0.84         412

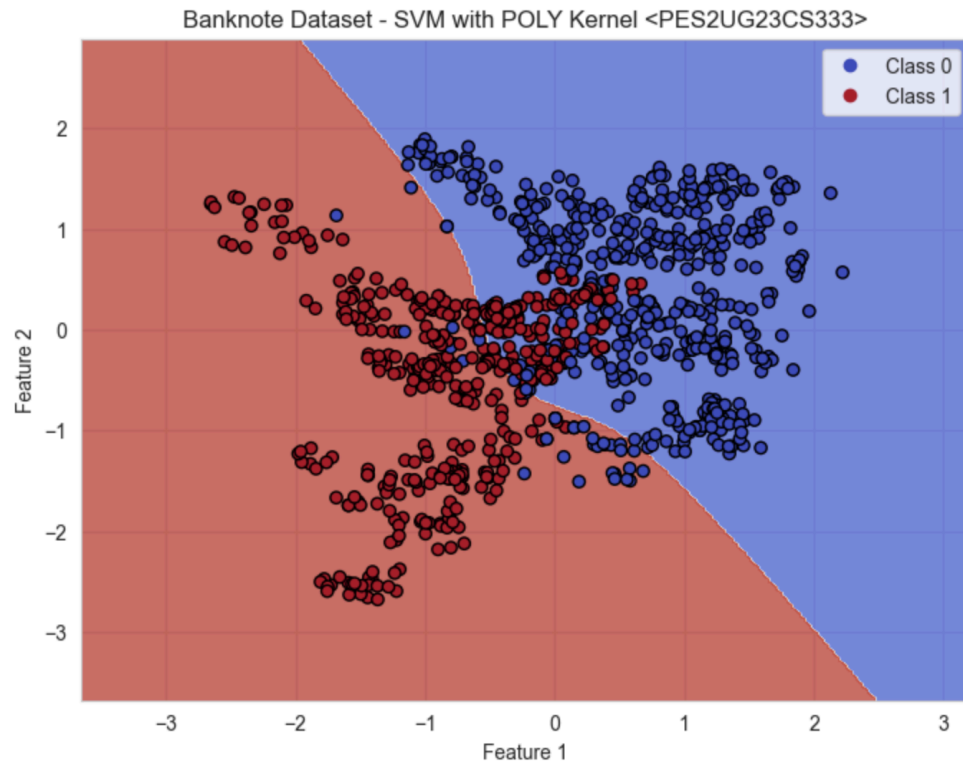
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```

Banknote Dataset - SVM with LINEAR Kernel <PES2UG23CS333>

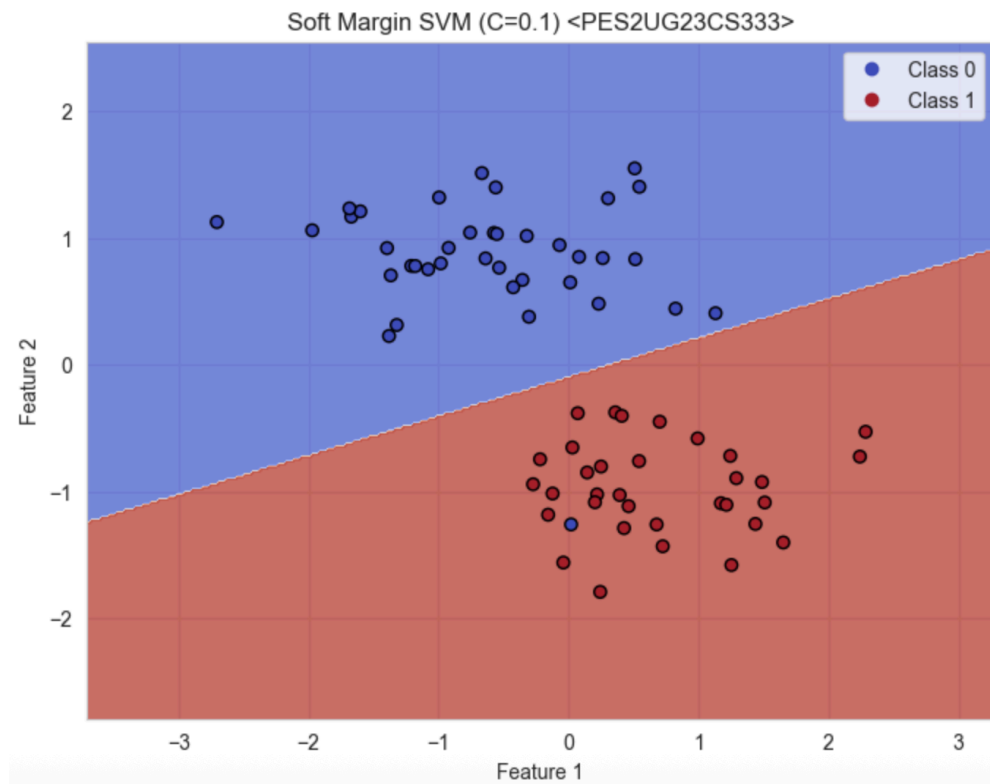


Banknote Dataset - SVM with RBF Kernel <PES2UG23CS333>





Hard and Soft Margin:



Hard Margin SVM (C=100) <PES2UG23CS333>

